

MATH-334 Numerical Analysis

Numerical Differentiation

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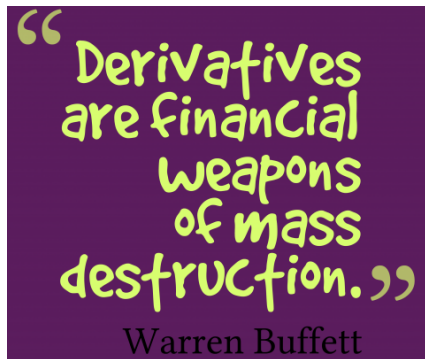
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 - Forward Difference Appr. of 1^{st} Derivative
 - Backward Difference Appr. of 1^{st} Derivative
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Calculus is the mathematics of change.

Standing at the heart of calculus are the related mathematical concepts of differentiation and integration.



Applications of Derivatives in Various fields/Sciences:

Such as in:

- Physics
- Biology
- Economics
- Chemistry
- Mathematics
- Others(Psychology, sociology & geology)

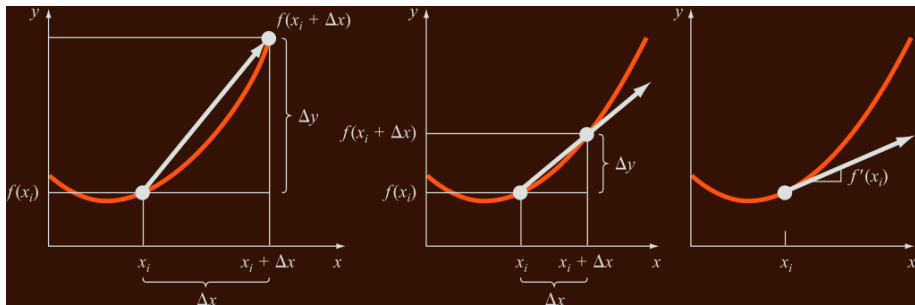


Derivative

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x_i + \Delta x) - f(x_i)}{\Delta x}$$

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The Taylor Theorem and Series

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Taylor series The Taylor series provides a means to express this idea mathematically.

$$f(x) = f(x_0) + \frac{(x - x_0)}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + R_n$$

In general, the n^{th} order Taylor series expansion will be exact for an n^{th} order polynomial. In other cases, the remainder term R_n is of the order of h^{n+1} , meaning:

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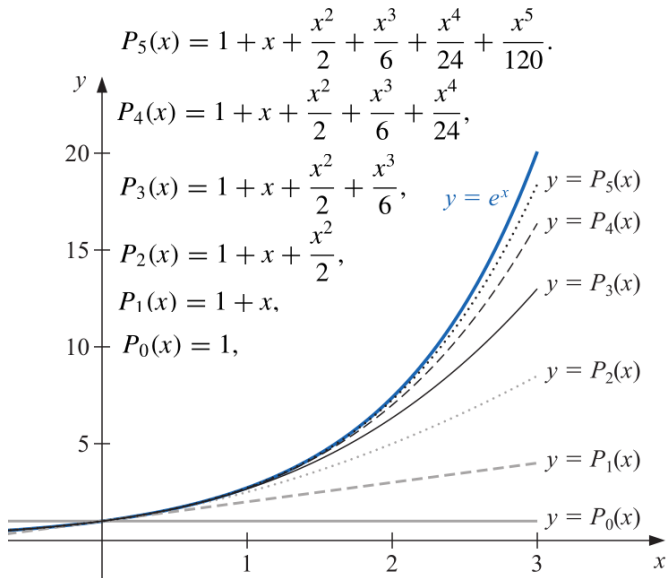
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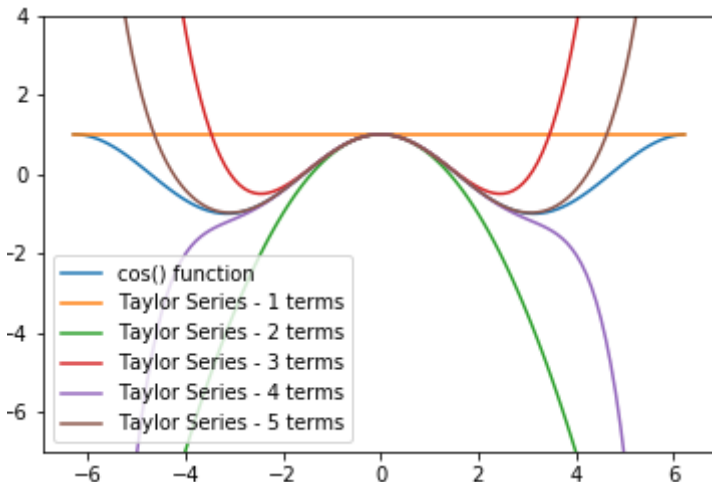
In general, the n^{th} order Taylor series expansion will be exact for an n^{th} order polynomial. In other cases, the remainder term R_n is of the order of h^{n+1} , meaning:

- The more terms are used, the smaller the error, and
- The smaller the spacing, the smaller the error for a given number of terms.

Approximation of e^x using Taylor's Series

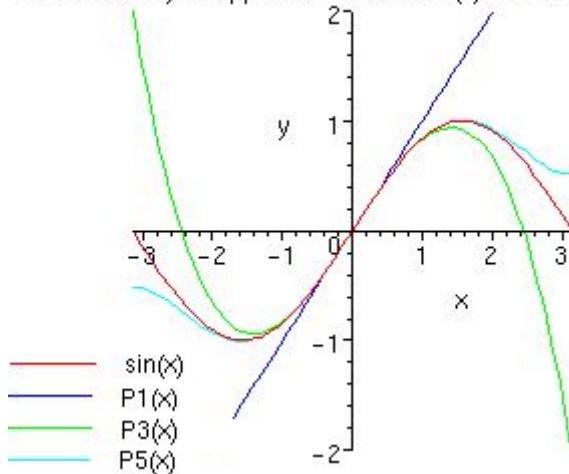


Approximation of $\cos x$ using Taylor's Series



Approximation of $\sin x$ using Taylor's Series

Low order Taylor approximations to $\sin(x)$ near $x=0$



Forward Difference Approximation of First Derivative Using Taylor Series

$$f(x) = f(x_0) + \frac{(x - x_0)}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + R_n$$

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It may be represented in the following form as well

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots + \frac{f^{(n)}(x_i)h^n}{n!} + \cdots$$

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From above equation we can write

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - \mathbf{O}(h)$$

Step Size

h is called the **step size**, that is, the length of the interval over which the approximation is made.

Also note that $\mathbf{O}(h)$ is error term and it depends upon step size.

$$f'(x_i) \simeq \frac{f(x_{i+1}) - f(x_i)}{h}$$

Derivative of a function may be written as

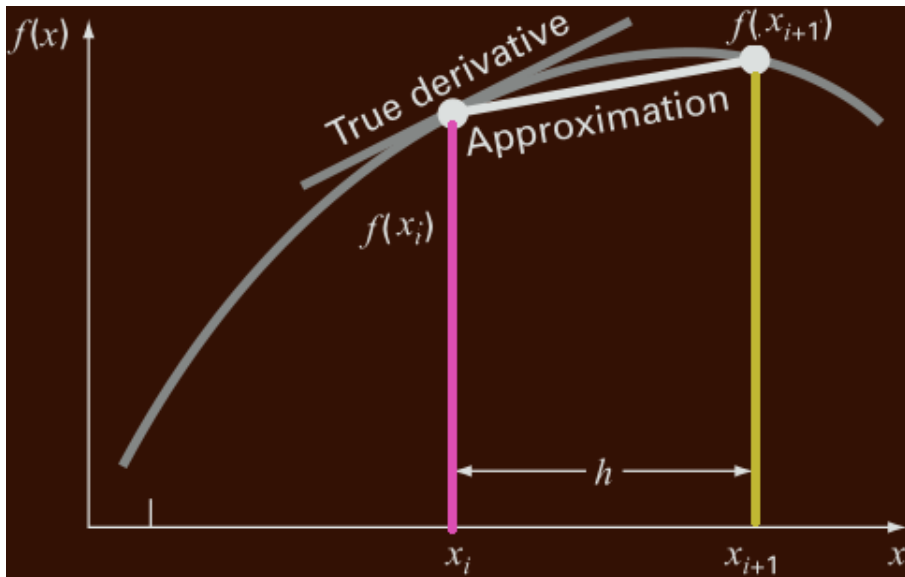
$$f'(x_i) \simeq \frac{\Delta f(x_i)}{h}$$

where $\Delta f(x_i)$ is referred to as the first forward difference.

$$\Delta f(x_i) = f(x_{i+1}) - f(x_i) \quad \text{Forward Difference Operator(FDO)}$$

It is termed as “forward” difference because it utilizes data at i and $i + 1$ to estimate the derivative.

The entire term $\frac{\Delta f(x_i)}{h}$ is referred to as a **first finite divided difference**.



Backward Difference Appr. of 1st Derivative

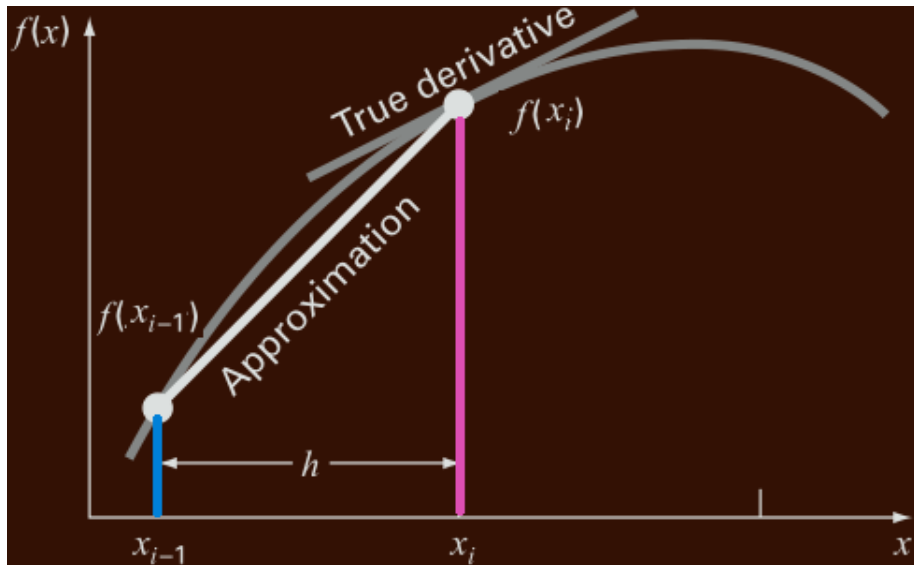
The Taylor series can be expanded backward to calculate a previous value on the basis of a present value

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \dots$$

From above equation we can write

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + \mathbf{O}(h)$$

where the error is $\mathbf{O}(h)$, and $\nabla f(x_i) = f(x_i) - f(x_{i-1})$ is referred to as the **first backward difference**.



Centered Difference Appr. of 1st Derivative

Taylor series expansion of $f(x_{i+1})$ and $f(x_{i-1})$ are given by

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \dots$$

So $f(x_{i+1}) - f(x_{i-1})$ gives

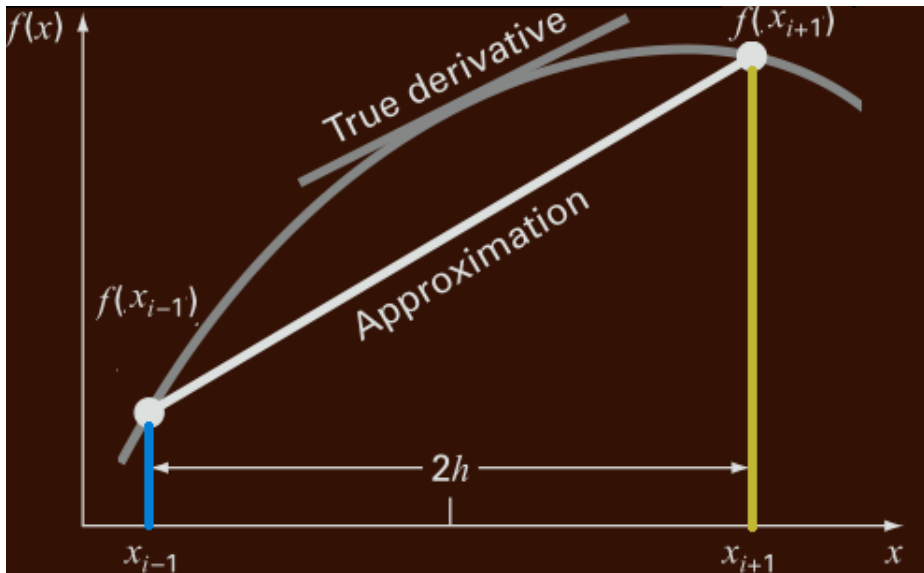
$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + \mathbf{O}(h^3)$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + \mathbf{O}(h^2)$$

$O(h^2)$ Notice that the truncation error is of the order of h^2 in contrast to the forward and backward approximations that were of the order of h .

More Accurate Taylor series analysis yields the practical information that the centered difference is a more accurate representation of the derivative

For example, if we halve the step size using a forward or backward difference, we would approximately halve the truncation error, whereas for the central difference, the error would be quartered



Approximation of 1st order Derivative

Forward Difference Formula

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

Backward Difference Formula

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$$

Central Difference Formula

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Example

Using a step size $h = 0.5$, estimate the first derivative by forward, backward difference and a centered difference approximation to of following, at $x = 0.5$,

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} \quad \text{or} \quad f'_i = \frac{f_{i+1} - f_i}{h} \quad (\text{FD})$$

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} \quad \text{or} \quad f'_i = \frac{f_i - f_{i-1}}{h} \quad (\text{BD})$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} \quad \text{or} \quad f'_i = \frac{f_{i+1} - f_{i-1}}{2h} \quad (\text{CD})$$

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We need to have values of $h = 0.5$, $x_i = 0.5$, x_{i-1} , x_{i+1} , $f(x_i)$, $f(x_{i-1})$ and $f(x_{i+1})$.

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$$f(x_{i-1}) = 1.2, \quad f(x_i) = 0.925, \quad f(x_{i+1}) = 0.2.$$

Using a step size $h = 0.5$, estimate the first derivative by forward, backward difference and a centered difference approximation to of following, at $x = 0.5$,
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$$f(x_{i-1}) = 1.2, f(x_i) = 0.925, f(x_{i+1}) = 0.2.$$

FD

$$\begin{aligned} f'(0.5) &= \frac{f_{i+1} - f_i}{h} \\ &= \frac{0.2 - 0.925}{0.5} \\ &= -1.45 \end{aligned}$$

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BD

$$\begin{aligned} f'(0.5) &= \frac{f_i - f_{i-1}}{h} \\ &= \frac{0.925 - 1.2}{0.5} \\ &= -0.55 \end{aligned}$$

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 $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

We need to have values of $h = 0.5$, $x_i = 0.5$, x_{i-1} , x_{i+1} , $f(x_i)$, $f(x_{i-1})$ and $f(x_{i+1})$. Thus $x_{i\pm 1} = x_i \pm h \implies x_{i+1} = 1$, $x_{i-1} = 0$

$$f(x_{i-1}) = 1.2, f(x_i) = 0.925, f(x_{i+1}) = 0.2.$$

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CD

$$\begin{aligned} f'(0.5) &= \frac{f_{i+1} - f_{i-1}}{2h} \\ &= \frac{0.2 - 1.2}{1} \\ &= -1 \end{aligned}$$

For $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

$f'(0.5) = -0.9125$. Accuracy in each case may be compared.

For $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

$f'(0.5) = -0.9125$. Accuracy in each case may be compared.

FD

$$f'(0.5) = \frac{f_{i+1} - f_i}{h}$$

$$= \frac{0.2 - 0.925}{0.5}$$

$$= -1.45$$

$$|\epsilon_t| =$$

$$100 \cdot \frac{-0.9125 + 1.45}{-0.9125}$$

$$|\epsilon_t| = 58.9\%$$

For $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

$f'(0.5) = -0.9125$. Accuracy in each case may be compared.

FD

$$\begin{aligned}
 f'(0.5) &= \frac{f_{i+1} - f_i}{h} \\
 &= \frac{0.2 - 0.925}{0.5} \\
 &= -1.45 \\
 |\epsilon_t| &= \\
 100 \cdot \frac{-0.9125 + 1.45}{-0.9125} \\
 |\epsilon_t| &= 58.9\%
 \end{aligned}$$

BD

$$\begin{aligned}
 f'(0.5) &= \frac{f_i - f_{i-1}}{h} \\
 &= \frac{0.925 - 1.2}{0.5} \\
 &= -0.55 \\
 |\epsilon_t| &= 39.7\%
 \end{aligned}$$

CD

$$\begin{aligned}
 f'(0.5) &= \frac{f_{i+1} - f_{i-1}}{2h} \\
 &= \frac{0.2 - 1.2}{1} \\
 &= -1 \\
 |\epsilon_t| &= 9.6\%
 \end{aligned}$$

Compare results, using a step size $h = 0.25$, estimate the first derivative by forward, backward difference and a centered difference approximation to of following, at $x = 0.5$,

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

Three approximations to the derivative $f'(a)$ are

1. the one-sided (forward) difference $\frac{f(a+h) - f(a)}{h}$
2. the one-sided (backward) difference $\frac{f(a) - f(a-h)}{h}$
3. the central difference $\frac{f(a+h) - f(a-h)}{2h}$

Example

Use a forward difference, and the values of h shown, to approximate the derivative of $\cos(x)$ at $x = \pi/3$.

$$(a) \ h = 0.1 \quad (b) \ h = 0.01 \quad (c) \ h = 0.001 \quad (d) \ h = 0.0001$$

Work to 8 decimal places throughout.

Solution

$$\begin{aligned} (a) \ f'(a) &\approx \frac{\cos(a+h) - \cos(a)}{h} = \frac{0.41104381 - 0.5}{0.1} = -0.88956192 \\ (b) \ f'(a) &\approx \frac{\cos(a+h) - \cos(a)}{h} = \frac{0.49131489 - 0.5}{0.01} = -0.86851095 \\ (c) \ f'(a) &\approx \frac{\cos(a+h) - \cos(a)}{h} = \frac{0.49913372 - 0.5}{0.001} = -0.86627526 \\ (d) \ f'(a) &\approx \frac{\cos(a+h) - \cos(a)}{h} = \frac{0.49991339 - 0.5}{0.0001} = -0.86605040 \end{aligned}$$

Example

Use **centered** difference, and the values of h shown, to approximate the derivative of $\cos(x)$ at $x = \pi/3$.

$$(a) h = 0.1 \quad (b) h = 0.01 \quad (c) h = 0.001 \quad (d) h = 0.0001$$

Work to 8 decimal places throughout.

Solution

$$\begin{aligned} (a) f'(a) &\approx \frac{\cos(a+h) - \cos(a-h)}{2h} = \frac{0.41104381 - 0.58396036}{0.2} = -0.86458275 \\ (b) f'(a) &\approx \frac{\cos(a+h) - \cos(a-h)}{2h} = \frac{0.49131489 - 0.50863511}{0.02} = -0.86601097 \\ (c) f'(a) &\approx \frac{\cos(a+h) - \cos(a-h)}{2h} = \frac{0.49913372 - 0.50086578}{0.002} = -0.86602526 \\ (d) f'(a) &\approx \frac{\cos(a+h) - \cos(a-h)}{2h} = \frac{0.49991339 - 0.50008660}{0.0002} = -0.86602540 \end{aligned}$$

Solve this

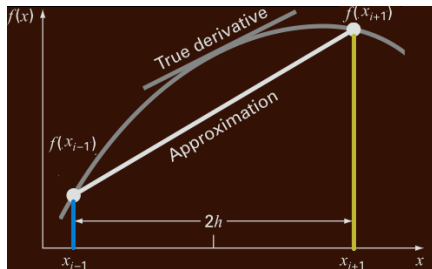
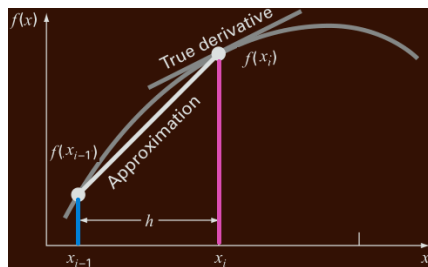
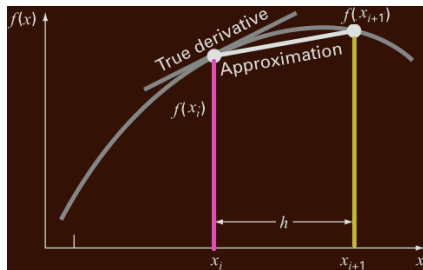
The distance x of a runner from a fixed point is measured (in metres) at intervals of half a second. The data obtained are

t	0.0	0.5	1.0	1.5	2.0
x	0.00	3.65	6.80	9.90	12.15

Use central differences to approximate the runner's velocity at times $t = 0.5$ s and $t = 1.25$ s.

Summary

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$$f'(x_i) \simeq \frac{f(x_{i+1}) - f(x_i)}{h}$$

$$f'(x_i) \simeq \frac{f(x_i) - f(x_{i-1}))}{h}$$

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